



# MASTER ROADMAP (DAY-WISE)

## Target:

By Sept 2026:

- Strong AI Infra / MLOps Engineer
- 18–30+ LPA target
- Potential 30–50+ LPA with OSS + strong projects

Since you're already a DevOps engineer, this roadmap is optimized for:

- ✓ Working professional
  - ✓ 3–4 hrs weekdays
  - ✓ 6–8 hrs weekends
  - ✓ Real projects over tutorials
- 



## DAILY SCHEDULE TEMPLATE

### Weekdays (Mon–Fri)



#### Morning (6:30–8:00 AM)

- 1 hr learning
  - 30 min revision/notes
- 



#### Evening (7:30–10:30 PM)

- 1 hr theory
  - 2 hrs hands-on/project
- 



## WEEKEND TEMPLATE

## Saturday

- Deep learning + implementation

## Sunday

- Project + revision + deployment
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# PHASE 1 — SYSTEMS + BACKEND + DEVOPS



## Weeks 1–8

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## WEEK 1 — Linux + Networking + Bash

### Day 1

- Linux filesystem
- pwd, ls, cd, cp, mv, rm
- permissions

Hands-on:

- Create directories/files
  - chmod/chown practice
- 

### Day 2

- grep
- awk
- sed

- pipes

Hands-on:

- Parse logs
- 

## Day 3

- Processes
- ps
- top
- htop
- kill
- systemctl

Hands-on:

- Monitor services
- 

## Day 4

- Networking basics
- ports
- DNS
- HTTP/HTTPS
- curl

Hands-on:

- curl APIs
  - check ports
- 

## Day 5

- Nginx basics
- reverse proxy

Hands-on:

- Setup Nginx locally
- 

## Day 6

PROJECT:

Deploy static website with:

- Nginx
  - SSL
  - reverse proxy
- 

## Day 7

REVISION + BLOG WRITING

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# WEEK 2 — Python Backend Foundations

## Day 8

- Python refresh
  - functions
  - classes
- 

## Day 9

- virtualenv
  - pip
  - requirements
-

## Day 10

- FastAPI intro

Hands-on:

- hello world API
- 

## Day 11

- Request/response
  - JSON
  - routing
- 

## Day 12

- PostgreSQL basics

Hands-on:

- CRUD
- 

## Day 13

PROJECT:

Build TODO API with:

- FastAPI
  - PostgreSQL
- 

## Day 14

Add:

- validation

- error handling
  - logging
- 

## **WEEK 3 — Docker**

### **Day 15**

- Docker basics
  - images
  - containers
- 

### **Day 16**

- Dockerfile
- layers

Hands-on:

- containerize FastAPI app
- 

### **Day 17**

- volumes
  - networking
- 

### **Day 18**

- Docker Compose
- 

### **Day 19**

- Multi-stage builds
- 

## Day 20

PROJECT:

Dockerize:

- app
  - postgres
  - redis
- 

## Day 21

Deploy locally fully containerized stack

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# WEEK 4 — Kubernetes Basics

## Day 22

- K8s architecture
  - pods
- 

## Day 23

- deployments
  - services
- 

## Day 24

- ingress

- namespaces
- 

## Day 25

- ConfigMaps
  - Secrets
- 

## Day 26

- Helm basics
- 

## Day 27

PROJECT:  
Deploy FastAPI app on K8s

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## Day 28

Add:

- autoscaling
  - ingress
  - monitoring
- 



## WEEK 5 — CI/CD + Terraform

## Day 29

- GitHub Actions basics
-



## Day 30

- CI pipelines
- 

## Day 31

- CD pipelines
- 

## Day 32

- Terraform basics
- 

## Day 33

- AWS infra provisioning
- 

## Day 34

PROJECT:

Auto deploy app using:

- GitHub Actions
  - Docker
  - Kubernetes
- 

## Day 35

REVISION + DOCUMENTATION

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## PHASE 2 — AI + LLMOPS



### Weeks 6–12

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### WEEK 6 — LLM Foundations

#### Day 36

- What are transformers
- 

#### Day 37

- Attention mechanism
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#### Day 38

- Embeddings
- 

#### Day 39

- Tokenization
- 

#### Day 40

- Hugging Face basics
-

## Day 41

Hands-on:

- run local models
- 

## Day 42

PROJECT:

Simple chatbot API

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# WEEK 7 — RAG + Vector DB

## Day 43

- RAG architecture
- 

## Day 44

- Chunking
- 

## Day 45

- Embeddings pipeline
- 

## Day 46

- Vector DB intro

Use:

- Qdrant
- 

## Day 47

- Retrieval pipelines
- 

## Day 48

PROJECT:  
PDF Chat System

Upload PDF → Ask questions

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## Day 49

Add:

- citations
  - metadata filtering
- 

# WEEK 8 — Observability + AI Monitoring

## Day 50

- OpenTelemetry basics
- 

## Day 51

- tracing

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## Day 52

- LangSmith/Phoenix
- 

## Day 53

- Prompt logging
- 

## Day 54

- Token tracking
- 

## Day 55

PROJECT:  
LLM monitoring dashboard

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## Day 56

Deploy dashboard

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# WEEK 9 — Agentic AI

## Day 57

- Agents from scratch
-

## Day 58

- Tool calling
- 

## Day 59

- Memory systems
- 

## Day 60

- Planning/reasoning
- 

## Day 61

- LangGraph/CrewAI
- 

## Day 62

PROJECT:  
Research agent

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## Day 63

Add:

- retries
  - memory
  - tracing
-



## WEEK 10 — MLflow + MLOps

### Day 64

- MLflow basics
- 

### Day 65

- experiment tracking
- 

### Day 66

- model registry
- 

### Day 67

- model deployment
- 

### Day 68

- inference pipelines
- 

### Day 69

PROJECT:  
ML deployment platform

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### Day 70

Add:

- monitoring
  - CI/CD
- 



## PHASE 3 — ELITE LEVEL



### Weeks 11–20

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## WEEK 11–12

### Distributed Systems

Learn:

- Redis
- Kafka
- queues
- pub/sub

PROJECT:

Event-driven AI pipeline

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## WEEK 13–14

### Advanced Kubernetes

Learn:

- operators
- service mesh
- scaling



PROJECT:  
Production AI serving platform

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## **WEEK 15–16**

### **GPU Serving + Optimization**

Learn:

- vLLM
- quantization
- batching

PROJECT:  
Optimized inference server

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## **WEEK 17–18**

### **Open Source Contributions**

Start contributing:

- LangChain
  - Qdrant
  - MLflow
  - OpenTelemetry
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## **WEEK 19–20**

### **System Design + Interview Prep**

Learn:

- AI infra architecture
  - scaling
  - multi-tenant systems
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## PROJECTS YOU MUST COMPLETE

### MUST HAVE

1. FastAPI production backend
  2. Kubernetes deployment
  3. Production RAG
  4. AI observability platform
  5. Multi-agent system
  6. ML deployment platform
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## WEEKLY OUTPUT RULE

Every week:

- ✓ 1 GitHub push/day
  - ✓ 1 LinkedIn post/week
  - ✓ 1 blog every 2 weeks
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## MOST IMPORTANT ADVICE

DO NOT:

- ✗ consume endless tutorials

DO:

- ✓ build
- ✓ deploy
- ✓ debug
- ✓ document

That's how high-paying engineers are created.

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# YOUR REALISTIC TIMELINE

## **By 3 Months**

- Better than average DevOps engineer

## **By 6 Months**

- Entry-level AI infra engineer

## **By 9–12 Months**

- Strong MLOps/LLMOps profile

## **By 12–18 Months**

- Competitive for high-paying AI infra roles